



ENHANCING SALES VISIBILITY AND INVENTORY DECISION-MAKING IN A SMALL RETAIL STATIONERY BUSINESS

BDM CAPSTONE PROJECT

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ABOUT THE BUSINESS AND OBJECTIVES



- Om Stationeries, Dattagalli, Mysuru
- Small neighbourhood retail shop
- 17 product categories, 664 SKUs
- Operations run entirely on owner experience – no structured records

PS1 – Profitability Visibility: The owner assumes a uniform 15–20% gross margin across all products, but has no structured way to verify this.

PS2 – Inventory Movement: Restocking relies on visual judgment, with no data to identify fast/slow-moving stock or capital lock-in.

PS3 – Experience vs. Data: All decisions rely on owner memory, with no records to verify or challenge them.

DATA COLLECTION

Sales Data:

- Store transactions recorded in a structured ledger and digitised into a spreadsheet
- Period: 10 Feb – 10 Mar 2026 (29 days, 458 transactions)

Inventory Data:

- Product list compiled with owner input; category mapping done using ChatGPT and verified
- Stock quantities based on owner estimates (14–16 Feb 2026)
- **No full physical count conducted to avoid operational disruption**

TOOLS

Python, Google Colab

Deduplication & Category Mapping

Unit Of Measurement Assignment

SKU Code Generation

Sales Processing

Error Handling

Incorrect date formats

→ standardised by script

Ambiguous product names

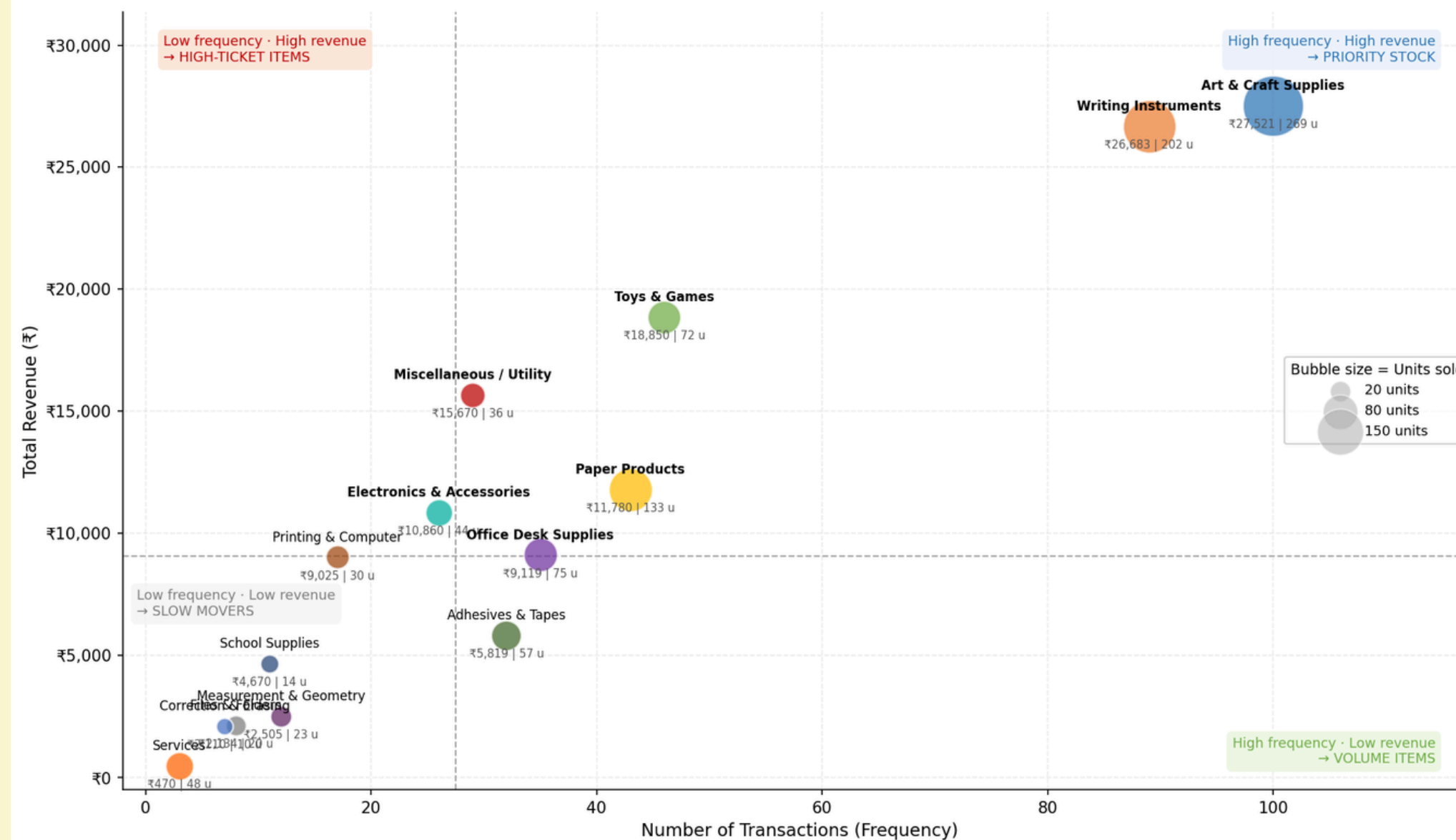
→ resolved via alias matching + manual check

Manual data risk

→ stock quantities are owner-stated; acknowledged as a limitation throughout



Fig 3.4: Category Performance — Transaction Frequency × Revenue × Units Sold
Each bubble = one category | Dashed lines = median frequency and median revenue

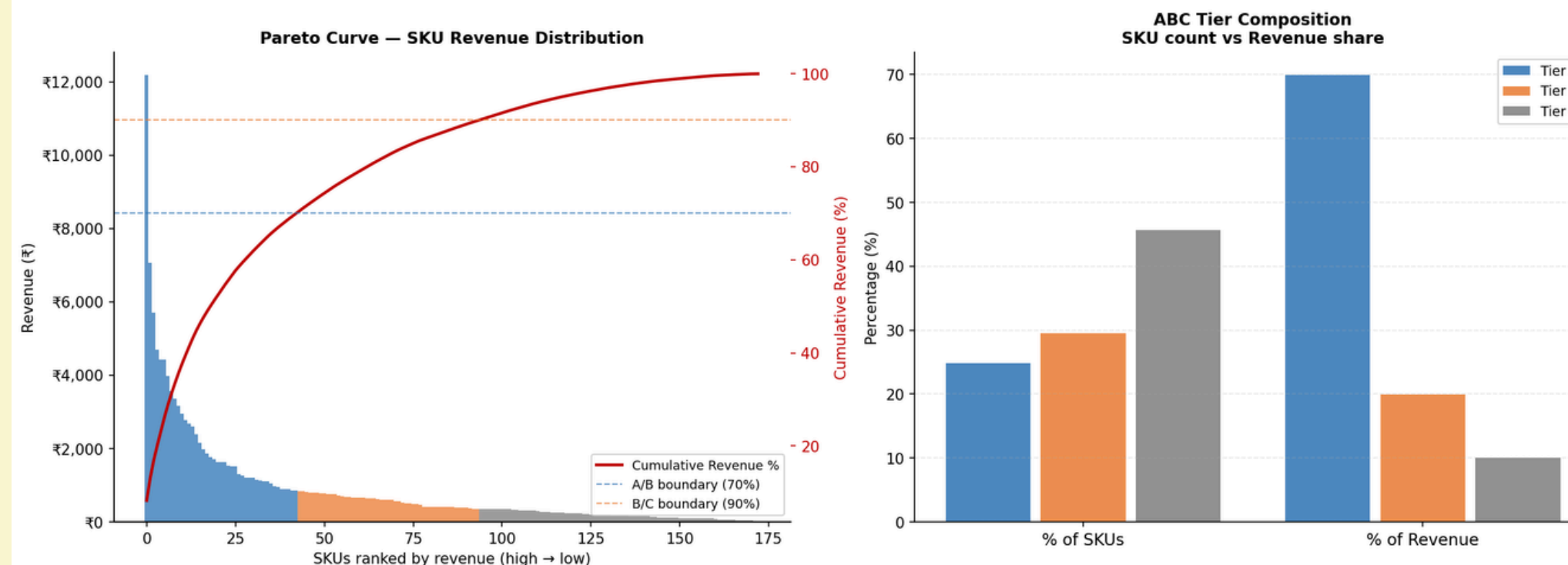


ANALYSIS



- Art & Craft and Writing Instruments top every metric – a stockout in either hits daily revenue directly
- 43 SKUs drive 70% of revenue; keep these in stock and the business is largely secured
- ₹3.8L sits in Writing Instruments moving at 7% velocity – capital is piled where sales are slow

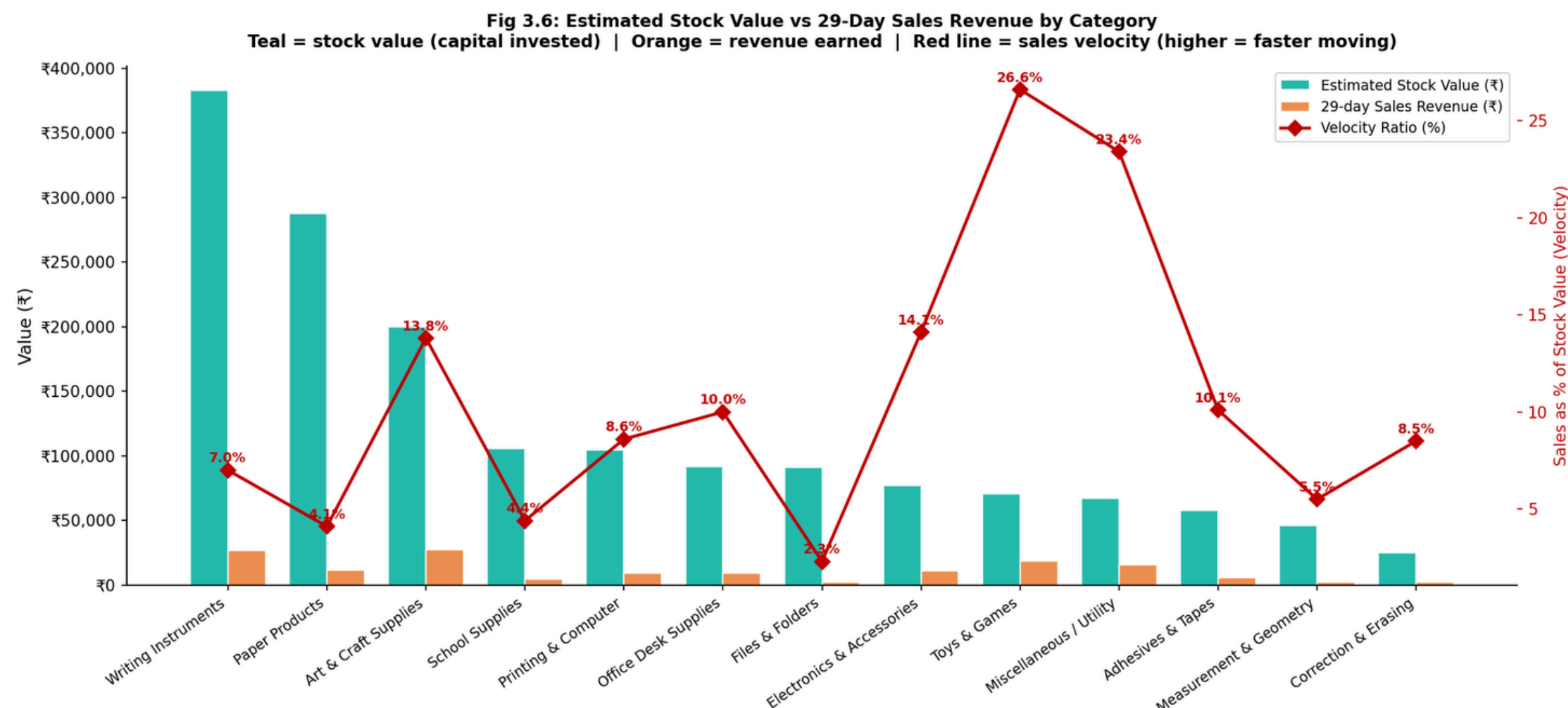
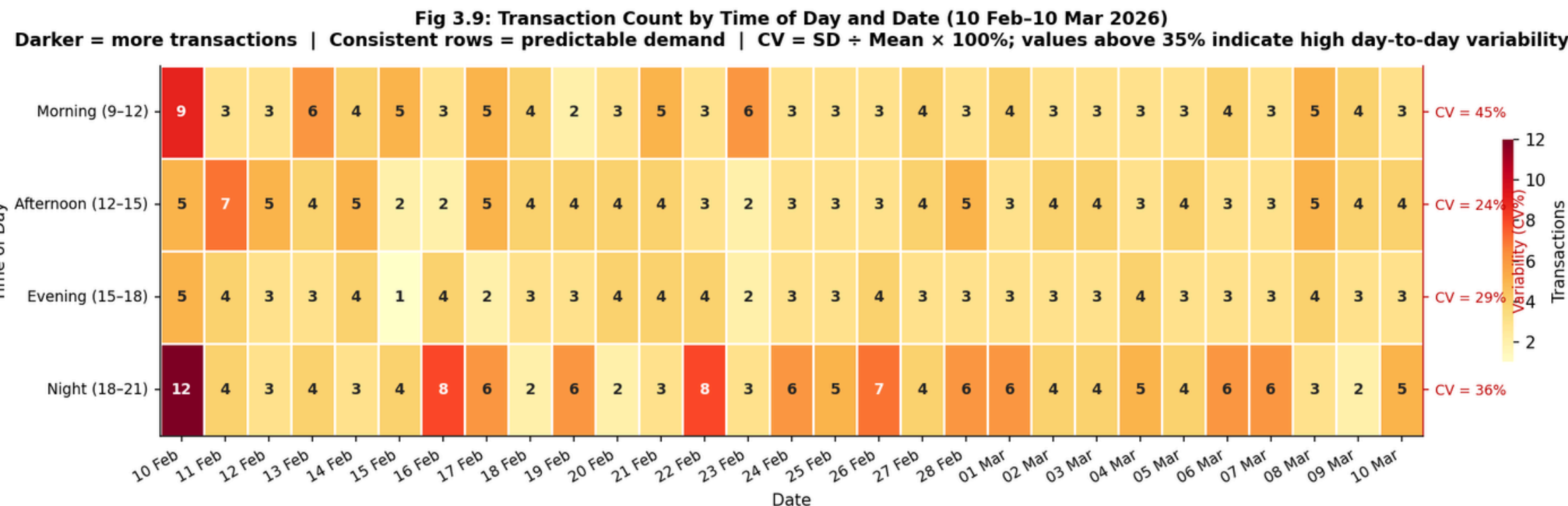
Fig 3.7: ABC Analysis — SKU Revenue Classification
Tier A = top 70% revenue | Tier B = next 20% | Tier C = bottom 10%





ANALYSIS

- Night and Morning are the busiest slots but also the most unpredictable – transaction counts swing wildly, making intuition-based prep unreliable
- Evening is the most stable window; anchor operational decisions here
- The store's top 3 best-sellers by volume are all priced below the margin floor – high sales is amplifying the loss, not offsetting it



INSIGHTS AND RECOMMENDATIONS



PS1 – Pricing

- Audit the categories bleeding margin first
- Stop assuming one margin floor fits all – set it at the category level
- Small price corrections on a few high-volume SKUs go a long way

PS2 – Inventory

- Not all products deserve equal attention – tier them and manage accordingly
- Some categories are sitting on months of unsold stock; that's working capital doing nothing

PS3 – Operations

- One small daily logging habit can replace years of guesswork
- Let the data tell you when to prepare more, not your gut

THANK YOU